

Based on the lecture notes provided in the sources, here is a comprehensive exam covering every section, including Multiple Choice Questions (MCQs) and potential Essay/Short Answer questions.

Pharmacology 1: Lecture 3 Examination

Part I: Multiple Choice Questions (MCQs)

Choose the most correct answer for each question.

Section 1: Introduction to Pharmacodynamics

1. Which of the following best defines "Pharmacodynamics"?

- ☐ A. What the body does to the drug.
- ☒ B. What the drug does to the body.
- ☐ C. The study of drug absorption and distribution.
- ☐ D. The process of drug excretion.

2. Pharmacodynamics includes the study of:

- ☐ A. Mechanism of action.
- ☐ B. Pharmacological actions.
- ☐ C. Adverse effects.
- ☒ D. All of the above.

Section 2: Types of Drug Action

3. Adrenaline increases the heart rate and force of contraction. This is an example of which type of drug action?

- ☒ A. Stimulation.
- ☐ B. Depression.
- ☐ C. Irritation.
- ☐ D. Replacement.

4. Barbiturates and general anesthetics are known to decrease the activity of specialized cells in the CNS. This action is classified as:

- A. Stimulation.
 - B. Depression.
 - C. Cytotoxic.
 - D. Replacement.
- 5. A "counterirritant" like Eucalyptus oil is used topically to relieve deep-seated pain. This falls under:
 - A. Stimulation.
 - B. Replacement.
 - C. Irritation.
 - D. Chemical action.
- 6. The use of Insulin in a patient with Diabetes Mellitus is an example of:
 - A. Cytotoxic action.
 - B. Stimulation.
 - C. Replacement.
 - D. Physical action.
- 7. Drugs that are selectively toxic to infecting organisms or cancer cells (like antibiotics) are categorized as:
 - A. Irritants.
 - B. Depressants.
 - C. Cytotoxic.
 - D. Osmotic.

Section 3: Non-receptor Mediated Mechanisms

- 8. 20% Mannitol is used in cerebral edema because it exerts an effect through:
 - A. Osmosis (Physical action).
 - B. Adsorption.
 - C. Enzyme inhibition.
 - D. Antibody production.

9. Which substance is used in drug poisoning because it acts by "adsorption"?

- A. Cough syrup.
- B. Antacids.
- C. Activated charcoal.
- D. Allopurinol.

10. Antacids treat peptic ulcers by neutralizing stomach acid. This is a:

- A. Physical action.
- B. Chemical action.
- C. Receptor-mediated action.
- D. Ion channel block.

11. Chelating agents are used to eliminate metals like iron and mercury by:

- A. Stimulating the heart.
- B. Trapping metals in a ring structure to form water-soluble complexes.
- C. Inhibiting xanthine oxidase.
- D. Blocking sodium channels.

12. Allopurinol reduces uric acid synthesis in chronic gout by inhibiting which enzyme?

- A. Serotonin transporter.
- B. Sodium channel.
- C. Xanthine oxidase.
- D. DNA polymerase.

13. Local anesthetics produce their effect by blocking:

- A. Antibody production.
- B. Nucleic acids.
- C. Sodium channels in the neuronal membrane.
- D. Serotonin reuptake.

14. Selective serotonin reuptake inhibitors (SSRIs) produce an antidepressant effect by binding to:

- A. Receptors.
- B. Transporters.
- C. Ion channels.
- D. Enzymes.

15. Vaccines like BCG work through which mechanism?

- A. Physical action.
- B. Cytotoxic action.
- C. Stimulating antibody production.
- D. Neutralization.

16. Cyclophosphamide is an anticancer drug that acts by binding to:

- A. Transporters.
- B. Enzymes.
- C. Nucleic acids.
- D. Antigens.

Section 4: Receptor-Mediated Mechanisms

17. Receptors are macromolecules that can be found in which of the following locations?

- A. Cell surface.
- B. Cytoplasm.
- C. Nucleus.
- D. All of the above.

18. Which of the following is an example of a receptor mentioned in the lecture?

- A. Adrenergic receptors.
- B. Cholinergic receptors.
- C. Opioid receptors.

- D. All of the above.

Part II: Essay & Short Answer Questions

Provide detailed explanations based on the lecture material.

1. Define Pharmacodynamics and list the three main areas it covers regarding drug study.
2. Compare and contrast "Stimulation" and "Depression" as types of drug action. Provide one example drug for each.
3. Explain the mechanism of "Physical Action" in drugs. List four different ways a drug can act physically and give an example for each.
4. Discuss how drugs act through chemical action, specifically focusing on antacids and chelating agents.
5. Describe the mechanism of action of Allopurinol in the treatment of gout.
6. What are the differences between how local anesthetics and SSRIs produce their effects? Mention the specific targets they bind to.
7. Define "Receptors" and explain the "Lock & Key" model of drug-receptor interaction based on the provided diagrams.

Answer Key

Part I: MCQs

1. B
2. D
3. A
4. B
5. C
6. C
7. C
8. A

- 9. C
- 10. B
- 11. B
- 12. C
- 13. C
- 14. B
- 15. C
- 16. C
- 17. D
- 18. D

Part II: Essay Hints (Summary of Source Info)

- 1. Definition: "What the drug does to the body". Three areas: Mechanism of action, pharmacological actions, and adverse effects.
- 2. Stimulation: Increases activity of specialized cells (e.g., Adrenaline). Depression: Decreases activity of specialized cells (e.g., Barbiturates).
- 3. Physical Action:
 - Osmosis: 20% Mannitol for edema.
 - Adsorption: Activated charcoal for poisoning.
 - Demulcent: Cough syrup coating mucosa.
 - Radioactivity: ¹³¹I for hyperthyroidism.
- 4. Chemical Action: Antacids (weak bases) neutralize stomach acid. Chelating agents trap metals in ring structures to form soluble complexes for excretion.
- 5. Allopurinol: Inhibits the enzyme xanthine oxidase, which prevents the oxidation of xanthine/hypoxanthine into uric acid.
- 6. Anesthetics vs SSRIs: Local anesthetics block sodium channels. SSRIs bind to 5-HT transporters to block serotonin reuptake.

7. Receptors: Macromolecules on cell surfaces, cytoplasm, or nucleus. They bind drugs to produce cellular changes, acting like a "Lock" that specific drug "Keys" fit into.